

1 Review: last 8 commits (17c5730..3dd0d60) — cbase2/ pipeline + DOCS_ASSESSMENT . . . v4

Scope note. Findings are pinned to 3dd0d60. During the review, two further commits appeared (86870e8, bb41bac, adding `cbase2/src/solvers.jl` and `JuMP/lpopt`); the solver section notes where they change the picture. Spot-checks I ran myself confirm the key findings below (noted inline).

1.1 1. Proposed closures vs. established CGE literature

Closure	Verdict
BF (geometric η)	Partially supported, over-claimed. Endpoints are BF-comparable: BF 2019 itself has a continuous specificity parameter <i>wg</i> and contrasts <i>full vs. no reallocation; the replication ships both variants. But the interpolation _between</i> them is project-specific — and the implemented middle also carries an ad hoc efficiency wedge (<code>mobile_labor.jl</code> :172–188) that <code>docs/labor_closures.md</code> :64–79 admits “is not established as the exact CES allocative-loss coefficient.” The Foundation I row omits this. $\eta=0$ is also not BF’s immobile closure: the code keeps one common wage with fixed quantities, while BF’s immobile case has sector-specific wages.
ALPHA (full-employment mobile)	Supported. This is the standard neoclassical/flexible-wage factor-market closure (LHR 2002 factor-market default; Dervis–de Melo–Robinson 1982). Minor: it is nested in BETA ($\eta_s=0$), not a parallel paradigm.
BETA (elastic labour supply)	Partially supported. A power-function supply curve is a defensible reduced form and directly answers R2, but the “labour–leisure interpretation” is not earned by the implementation (no income effect); the canonical references (e.g. Boeters–Savard 2011 on labour-market closures, wage curves) are absent, and R1.5’s wage-curve demand is a different family.
GAMMA (sticky real wage)	Loosely labelled, and its headline is stale. The canonical fixed-wage closure is <i>one-sided</i> (nominal floor) with rationed employment and involuntary unemployment (LHR pp. 9–10; DDR ch. 6). GAMMA is a two-sided real-wage peg with uncapped employment — i.e. perfectly elastic labour supply, which the doc itself identifies as BETA’s $\eta_s \rightarrow \infty$ limit. It therefore cannot be an independent paradigm. Worse: the “+19.3 pp at mult 10” headline comes from the retired unfinanced shock; the financed F2 run is $\approx$ employment-neutral (<code>process_comments.md</code> :122).
DELTA (IO end-point)	Right family, imprecise object. Robinson 2006 / Rose 1995 / McGregor–Swales–Yin 1996 support the equivalence claim, but it requires fixed prices of <i>all</i> inputs + unconstrained factor supplies. The implemented analytic counterpart (<code>closures.jl</code> :80–134) includes endogenous consumption, saving and import leaks — a Type II/SAM-style multiplier, not $(I-A)^{-1}$ or a Type I multiplier (R1.4’s unresolved point). The doc’s corrective “the IO multiplier requires sticky wages” is too narrow.

- **Limit relations:** BF $\eta=1 \square$ ALPHA is *definitional* in the code (true by construction, not a result); BETA $\eta_s \rightarrow \infty \approx$ GAMMA is correct (one-sided) — and is itself evidence GAMMA is not independent; GAMMA+Leontief = DELTA is effectively verified (`verify_v3.jl`:140–161).
- **Financing closures:** F1 is a demand-composition shifter, not a financing closure (the doc half-admits this); F2/F3 do map onto the standard government-budget and external closures of the CGE macro-closure taxonomy (LHR 2002, Table 3). The claim that debt finance “collapses into the tax variant” is correct only under unstated assumptions — static, closed, Ricardian, lump-sum, domestically held; it fails with hand-to-mouth households (which BF 2022 itself models), distortionary taxes, or foreign holders. The endogenous-money caveat is right, but “closest representation” framing is not literature-supported.
- **Coverage table defects:** says “four” BF elements but lists **six** rows; adjustment costs are BF **2019**, not 2022; the ZLB replication parenthetical is wrong (`VALIDATION.md` validates inflation and unemployment); the wage-rigidity row omits one-sidedness/rationing.
- **Missing canonical refs:** LHR 2002, Rattsø 1982, Robinson 1991, Taylor 1990, Sen 1963, Boeters–Savard 2011, Miller–Blair (Type I/II).

1.2 2. Concrete implementation bugs

1. **CRITICAL — the mobile system enforces only N–1 market-clearing equations, and the doc says the opposite.** `mobile_labor.jl:320–329` drops the N-th clearing equation, with a comment stating it is Walras-redundant again (lines 248–253) — directly contradicting `DOCS_ASSESSMENT.md` (“ALL N clearing equations are enforced; the N-th market is not Walras-redundant once imports leak”). The omitted market is not zero in general: the p-weighted residual sum equals $S - (I+X-M)$, which the model never imposes. The canary at `verify_v3.jl:85–98` only *prints* (no assert) and reportedly showed $\text{diff} \approx -0.094$ at one equilibrium. A subagent constructed a one-sector counterexample where all residuals are 0 but goods-market excess demand is 0.1. Perfect residuals can certify a non-equilibrium.
2. **CRITICAL — drop_sectors breaks CES share normalization** (`calibration.jl:60–78`): rows of Ω no longer sum to 1 (0.971 for 70s; 0.549 for “reduced”). With all $p=1$, the intermediate price index at $\theta=0.999$ collapses to $2.2e-13$ vs. 1.0 at $\theta=1$ — an artificial technology discontinuity along the θ -continuation.
3. **MAJOR — mixed GDP normalizations after dropping sectors:** `labor_share` sums to 0.9878 (70s) / 0.8832 (reduced), not 1, while $E_{h0} = 1 - \Sigma g$ assumes baseline income 1 (`calibration.jl:155`). Calibrated s , taxes, shock vectors and the BETA anchor are inconsistent across dataset variants.
4. **MAJOR — production-side accounting doesn’t close for $\eta < 1$:** the zero-profit price uses cost-minimizing labour under the wedge while household income pays the *interpolated* labour (`mobile_labor.jl:282–315`); $p \cdot y - wL + p \cdot \text{int}$ by the wedge amount, which is distributed to no one.
5. **MAJOR — baseline “exact clearing” relies on clamping:** `c0_dom = max.(c0_dom, 0)` (`calibration.jl:144–151`) clamps negative household residuals in 33 sectors, total mass $\approx$ **0.0873 GDP units ($\approx 8.6\text{--}8.7\%$ of GDP)** on the working dataset; the claimed exact baseline is not an equilibrium, and s is driven by this adjustment.
6. **MAJOR — notebook 03 no longer runs** against the current kernel (unrecalibrated `household_baseline = zeros()` `DomainError`); its API calls and budget assertions are out of date.
7. **MAJOR — public_budget prices at $\text{sum}(\text{gov_demand})$ while taxes are $\text{dot}(p, \text{gov_demand})$** (`financing.jl:117–120, 150–153`); government budget reports fail as soon as $p \neq 1$ (≈ 0.214 model units at $p=2$).
8. **MAJOR — external_balance returns only the programme’s import content** (`financing.jl:155–168`), not the documented external balance; the “ $F = 0.00147$ ” figure is ambiguous ($\approx$ half of $\text{dot}(p, m \cdot g)$ at baseline).
9. **MAJOR — verify_v3.jl defects:** F1 tested with an all-ones (no-op) preference vector; the headline rows use $\theta=1.0$ while the reference `ref` is the final continuation point ($\theta=0.5$) — results mix financing effects with a technology change; import canary adds quantities to values; DELTA asserts $\text{rel} < 5e-3$ while the doc quotes 0.0; hardcoded `/workspace/...` paths; the BETA “identification” test is circular (below).
10. **MINOR/MAJOR — fixed-wage scale guard** (`mobile_labor.jl:466–473`) is type-based, not rank-based: it throws on determinate open economies with `NoFinancing()` yet is bypassed by wrapping a zero vector in `ExternalDebt`.

1.3 3. Mathematical/conceptual mistakes

1. **The “corrected equilibrium formulation” is wrong as documented (critical).** The doc’s Walras claim is false for the mobile system (see bug 1), and it is internally contradictory: the code comment says the N-th equation is redundant again. Mathematically, the dropped residual equals the unmodeled external-account imbalance. The two wage regimes therefore solve *different models* (fixed: all N clearing; mobile: N–1 clearing + CPI). Fix: either impose the external-balance equation and keep N–1 clearing, or define imports/external balance residually and actually assert the canary.
2. **The saving-rate story is tautological and stale (major).** s is definitionally the residual of the baseline clearing equation, not a quantity “forced” by the identity — and with the clamp active the identity $S = I+X-M$ fails by exactly the clamped mass. The headline numbers ($s=0.398$, $\tau_0=0.214$, $X=0.422$, $I=0.162$) are the **pre-drop 71-sector** values; the working 70-sector pipeline prints **$s=0.4259$, $\tau_0=0.2167$, $X=0.4271$, $I=0.1644$** .
3. **BF closure (major):** interpolation limits hold, but $\eta < 1$ violates zero-profit/budget consistency (bug 4); the wedge curvature is admittedly not derived; η identifies no structural friction parameter (it is an exogenous interpolation weight), and $\eta=0$ is mislabelled as BF’s immobile benchmark.
4. **BETA (major/unproven):** the “identification test” $\log(L/1)/\log(w/1)$ recovers η s *by construction* (it is the enforced residual, with w_0 hardcoded to 1 and $L=1$ only on full data); the anchor $w_0=1$ contradicts the doc’s own open item that the baseline wage reads ≈ 0.54 ; the fold fallback (`closures.jl:171–190`) can jump branches.
5. **DELTA exactness (major, claim correct but for the wrong reason):** with a demand-only shock, w fixed and $A=1$, **$p=1$ is an exact equilibrium for all elasticities**, so the analytic system matches regardless of the $\varepsilon \rightarrow 0$ limit; the limit is about uniqueness/branch stability, not the equality. The quoted $L = 1.1656/1.1767$ are pre-drop 71-sector values (70s $\approx 1.1365/1.1469$), and the near-unit round-gain column sums (≈ 0.976) make the exactness fragile.
6. **Tax wording contradiction (major):** the same section calls F1/F3’s tax “fixed nominal $T = \Sigma gG$ ” and then correctly describes $T = p'gG$; the code implements the latter (price-indexed real tax). Also, $\text{CPI}=1$ exists only in the mobile

system — the fixed/DELTA rows are anchored by $w=1$.

7. **Solver diagnosis (minor–major):** the doc conflates ForwardDiff (AD) with finite differences, and “kinked residuals under ForwardDiff, not model structure” is contradicted by reproduction (below).
8. **Misc:** `exo_scale` continuation walks a *family of recalibrated economies* (s rises 0–0.43), so the mechanical Törnqvist decline is built in and the doc presents it only as a solver device; “Törnqvist real GDP” is a household-consumption index, not GDP; the asserted budget identity $\Sigma pc = (1-s)E$ is definitional and validates nothing.

1.4 4. Is the proposed solver strategy feasible?

Partially — and its central diagnostic claim is wrong.

- **What actually runs:** the mobile system has **141 unknowns** (not 140), solved by NonlinearSolve’s default polyalg (Broyden, Klement, Newton–Raphson, TrustRegion, LM last), with AD (`AutoForwardDiff`), not FD, despite the doc’s “FD-Newton”. Tolerances: core 1e-6; `solver_sweep.jl:39` uses 1e-8, against the project’s own recorded lesson “do not tighten tolerances — tighter settings stall.”
- **The stall reproduces and is not a solver-setup artifact:** at the failing rung, Newton-AD stalls at init; Newton-FD can’t reduce it; **LM-AD and LM-FD both land on exactly $3.612e-4$** ; AD and FD Jacobians agree to $\sim 4e-10$; the clamp/floor kinks are inactive at the stall. The Jacobian has **$\text{cond} \approx 3.4e10$** with $\sigma_{\min} \approx 9.9e-6$, and **99.3% of the residual lies along the $\sigma_{\min}$ direction** (a near-flat price of tiny sector 70). That is ill-conditioning/near-singularity — possibly a fold, i.e. *_model/calibration structure* — not init-luck fixable by swapping algorithms.
- **Runtime claim:** the default path takes **4–7 minutes per attempt**, not “seconds” (5.4 s holds only for direct Newton/LM calls). The proposed sweep as written is multi-hour, runs at the wrong rung (target `exo=1` instead of the failing `esc0`), and lacks the FD/LM arms and conditioning diagnostics.
- **IPOPT/JuMP:** technically sound for the *square* system (the B&F MATLAB code indeed calls KNITRO/fmincon on a zero-objective feasibility NLP — the analogy is accurate), provided the residual gate is kept (a KKT point of $\min \frac{1}{2}\|r\|^2$ is not a root) and one budgets for dense-FD cost (≈ 282 residual calls ≈ 6 s per Jacobian refresh). It is **not** a complementarity solver: Closure C (0 I–L w/P– 0) needs an MPEC/smoothing/PATH-style formulation.
- **The two commits added during review** implement exactly this A–D ladder (`cbase2/src/solvers.jl`), which is the right direction — but the LM arm still contains `LevenbergMarquardt(; ad = AutoFiniteDiff())` (line 194), an invalid keyword in the installed NonlinearSolve that a `try/catch` silently swallows into default AD. That bug persists at current HEAD and should be fixed before trusting any ladder results.

Recommended order: (1) scaling/conditioning experiment (quantities by λ , row equilibration; measure cond , $\sigma_{\min}$, residual alignment) on the failing rung at 1e-6; (2) reformulate the wedge/tiny-sector sensitivities with bounded derivatives; (3) fold detection along the p_{70} null direction; (4) only then redesign the init/algorithm sweep; (5) keep IPOPT as verification/polish, not as the primary fix — and fix the `ad` keyword first.

1.4.1 Bottom line

The v4 document’s closure *map* is a reasonable project-specific synthesis, but three claims do not survive contact with the code and literature: **(i)** the “ALL N clearing equations enforced” correction is false (and the omitted market is not redundant), which undermines the acceptance-test evidence; **(ii)** GAMMA is not the standard Keynesian closure and its headline number is from the retired unfinanced shock; **(iii)** the solver floor is an ill-conditioning/fold problem aligned with a near-singular direction, not an init/FD/AD setup issue. The BF interpolation and DELTA exactness are defensible only with qualifications the doc currently omits (endogenous wedge, Type II multiplier, $\eta=0 \neq$ BF-immobile). Highest-value fixes: renormalize/recalibrate after `drop_sectors`, disclose and reconcile the baseline clamp, impose or residually define the external account and assert it, correct the stale 71-sector headline numbers, and run the scaling experiment before any further solver work.